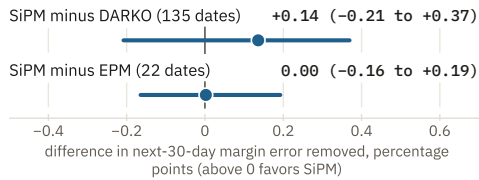


Running The Bulls, Ahead of the Herd

The NBA is driven by stars. How do the Chicago Bulls find the next star and put the best supporting cast around him?

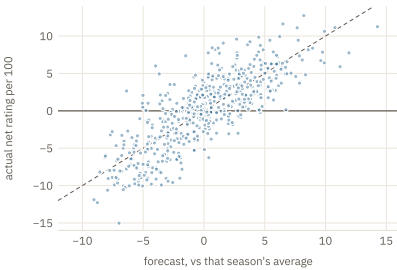
Winning in the NBA is a constant battle to scrap for every edge you can possibly find. This is traditionally thought of as the domain of analytics, which is often seen as being in tension with the eye test. In fact, the two are complementary.

An NBA analytics department must have an impact model that blends lineup ratings and box-score data. Mine is on par with the leading public models.



Ratings frozen on each 2025-26 game day, scored on the next 30 days of game margins. Dot: difference; bar: 90% interval.

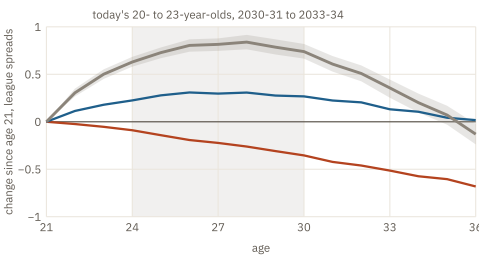
The forecast tracks team success



One dot per team-season, 2008-09 to 2025-26 (540): prior-season SiPM summed over the possessions each player went on to play. Correlation 0.80; last season's net rating alone gives 0.60.

The question is what you do from there. I want to model trajectories, test questions, and always work backwards from the end goal: winning NBA championships.

Overall impact (band: 90% interval) Explosive traits, average
Skill traits, average



Typical change since age 21, every regular season 1996-97 to 2025-26. Shaded: the ages today's 20- to 23-year-olds reach in 2030-31 to 2033-34.

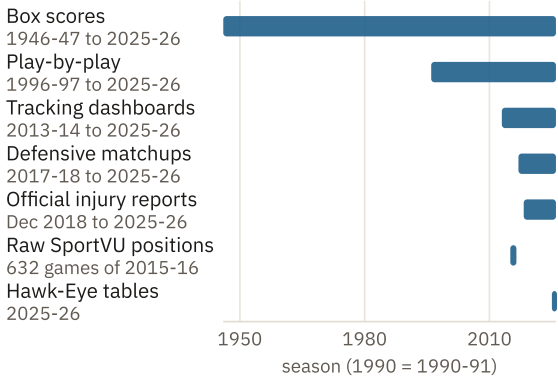
The NBA has singular star talents. Looking at the best players today, each has a unique game:

Wemby Jokic SGA Giannis Kawhi LeBron Steph KD

The name came from my hypothesis that was found to be unproven: that identifying the next singular star could be done by data alone. Data narrows the list, shows whose game is still growing, and finds who fits around him. Singularity is identified by watching the game.

Analytics work is only as good as the underlying data.

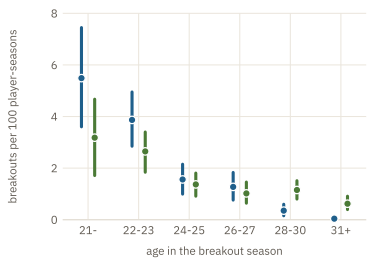
Seasons covered, by source



The seasons each public data source covers (tracking dashboards are NBA.com's).

Even the official sources contain errors. The audit found 49 root causes of errors in the providers' own data. The most were in box scores (14), tracking dashboards (13) and draft and combine (6). Each one is corrected, kept out of the analysis, or flagged where it is used. [Every error, by source.](#)

Offensive breakouts Defensive breakouts



Breakouts (a top-2.5% one-season rise against a player's own past), 1998-99 to 2025-26. Defense breaks out about two years later.

1 The model

SiPM (Singular Plus-Minus) rates how many points per 100 possessions a player adds, built from public box-score, play-by-play and lineup data. Its forecasts are scored on seasons it never saw. With ratings frozen on every game day of 2025-26 and every week of the two seasons before, it is level with two leading public impact models: DARKO in every replay, and EPM on every 2025-26 date with an archived copy.

0.80

correlation with next season's team net rating

0.98

slope: a point of forecast is a point of margin (90% interval 0.93 to 1.04)

0.60

the same correlation using last season's team net rating

Add up each player's prior-season SiPM, weighted by the minutes he went on to play, and you get the team's next-season net rating almost point for point. It uses the real minutes, so it tests the ratings, not a minutes forecast.

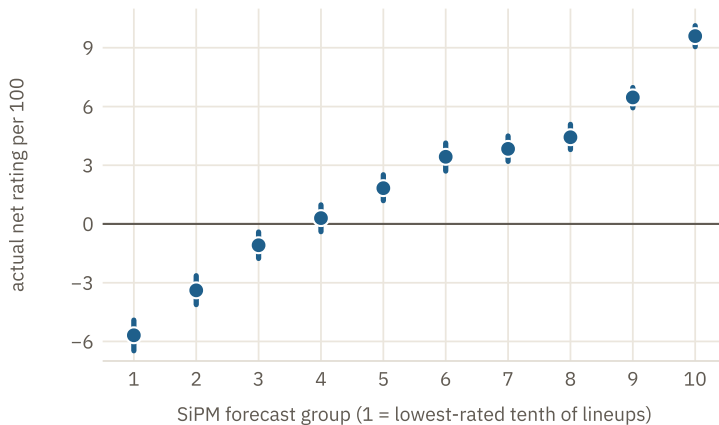
SiPM against two leading public models

Comparison and replay	Dates	Difference (90% interval)	Reading
SiPM minus DARKO every game day, 2025-26	135	+0.14 (-0.21 to +0.37)	level
SiPM minus DARKO every week, 2025-26	21	+0.17 (-0.16 to +0.35)	level
SiPM minus DARKO every week, 2024-25	21	-0.23 (-0.52 to +0.12)	level
SiPM minus DARKO every week, 2023-24	21	-0.09 (-0.47 to +0.22)	level
SiPM minus EPM game days with an archived EPM, 2025-26	22	0.00 (-0.16 to +0.19)	level
SiPM minus EPM weeks with an archived EPM, 2025-26	3	+0.07 (-0.12 to +0.25)	level

Score: how much of the error in forecasting the next 30 days of game margins each model's ratings remove, in percentage points; above 0 favors SiPM. "Level" means the interval includes 0.

FIGURE 1

The higher SiPM rates a lineup, the better it plays on seasons the model never saw



Average actual net rating climbs from the lowest-rated tenth of lineups to the highest. A single lineup stays noisy (correlation 0.29), so the forecast is strongest for groups of lineups and for roster decisions.

Dot: group average. Bar: 90% interval.

Like other impact models, it overvalues non-creating bigs and undervalues primary creators. Finding the underlying cause of these residuals, and framing the data and the analysis accordingly, is a primary focus of this work.

2 Findings

Findings at a glance

- Trajectories. Most explosive box-score traits are highest at 21 or younger and fade first; shooting and passing improve until 26 to 28, ball security until 25; defensive breakouts come about two years after offensive ones.
- Usage and efficiency. When a teammate's absence hands a player more of the ball, his true shooting falls about half a point per usage point. Most of that comes from losing the teammate who set him up; the extra shots themselves cost little. Comparing players hides the cost.
- Fit. Spacers and interior bigs add value next to a star creator. Two star creators together are worth about 0.9 per 100 less than their ratings add up to.
- Threes versus twos. In the half court, threes and twos are now worth the same once offensive rebounds and the opponent's next trip are counted: NBA teams have found the balance. Shots at the rim still beat a three; midrange twos do not. What I want to track is how that balance shifts, so the Bulls can lead the pack on the next play-style revolution.
- What data cannot see. On defense, award votes add signal the numbers miss. A model built on public data captures about half of the signal in a player's next three seasons.
- How it was tested. 591 questions tested against the record, many of them answered no. Most numbers carry a 90% interval.

Basketball minds are constantly asking how to play the game better. Those questions are testable, and testing them shows how well the eye test and analytics go together.

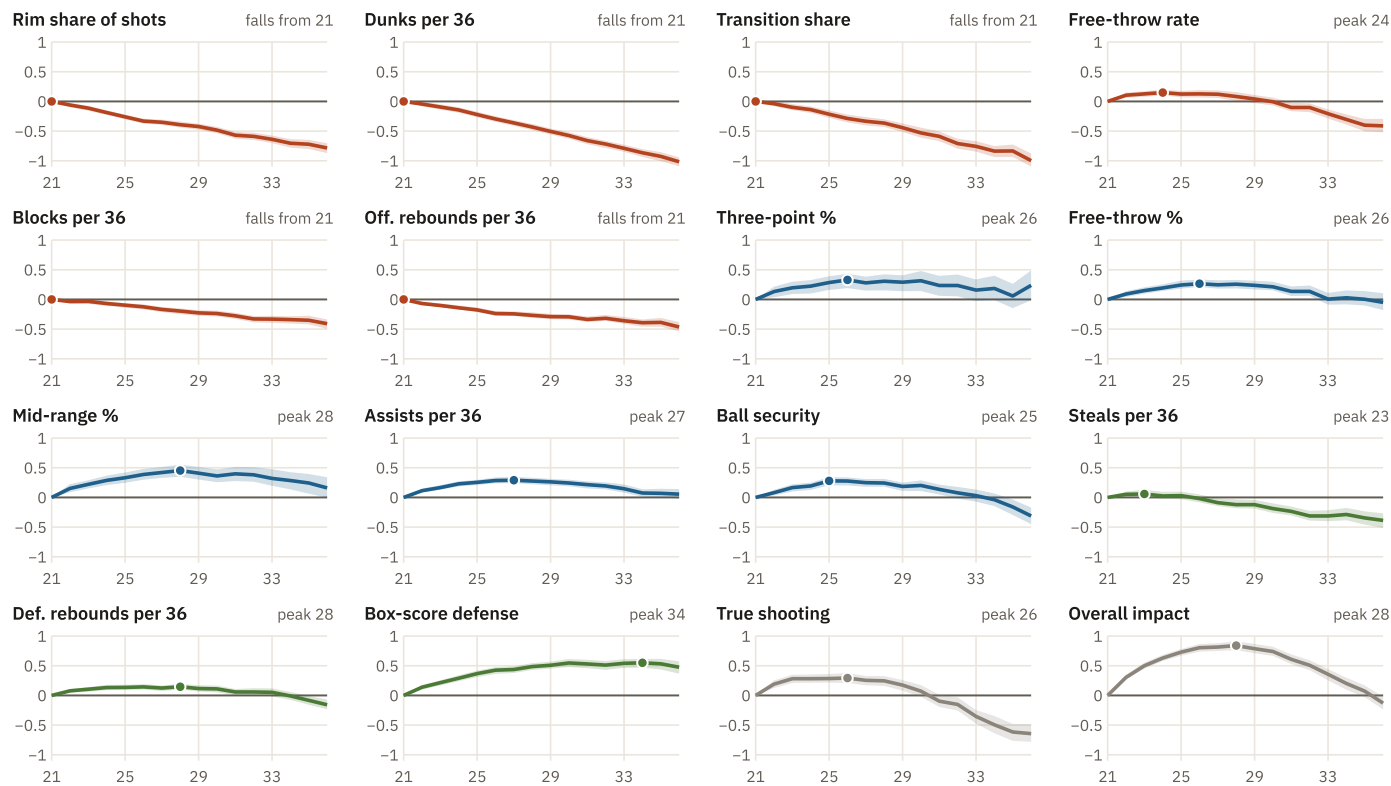
Trajectories: how each part of a player's game grows and fades

A rebuilding team invests in primes. These figures show which traits a roster gains and loses as its players age.

FIGURE 2

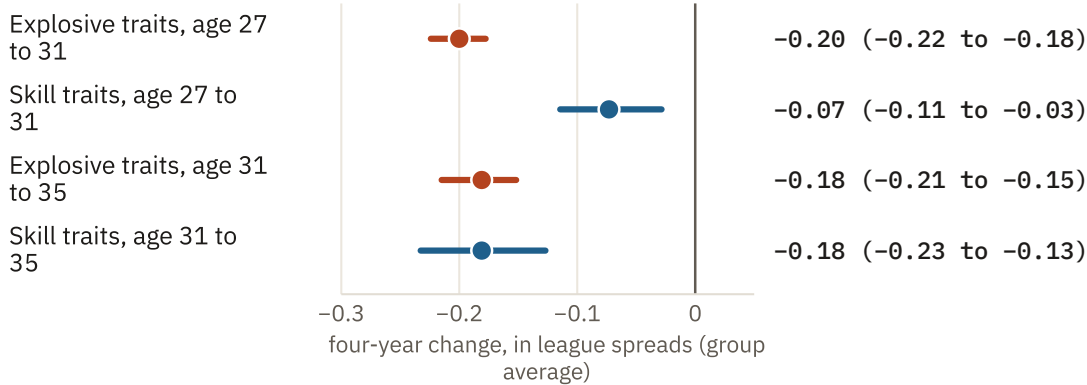
Most explosive box-score traits are highest at 21 or younger and fade first; shooting and passing improve until 26 to 28, ball security until 25

— Explosive traits — Skill traits — Defense — Overall



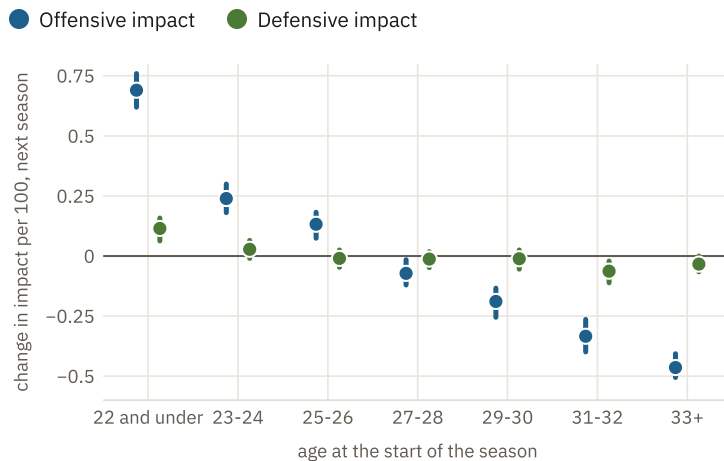
A roster's explosive strengths are the first to go. Rim shots, dunks, transition shots, blocks and offensive rebounds are highest at 21 or younger and slide from there; shooting and passing keep improving until about 26 to 28, ball security until 25, and overall impact peaks at 28.

Line: typical change since age 21, in league spreads (up = better, so ball security is turnovers turned upside down). Band: 90% interval. Dot: peak age. Box-score defense, the one trait that rises into the 30s, is the weakest measure here: age is one of its model's inputs, so its curve partly echoes that age term. Free-throw rate sits in the explosive group but peaks later than the rest.

FIGURE 3**From 27 to 31, explosive traits fall almost three times as fast as skills; after 31 both fall together**

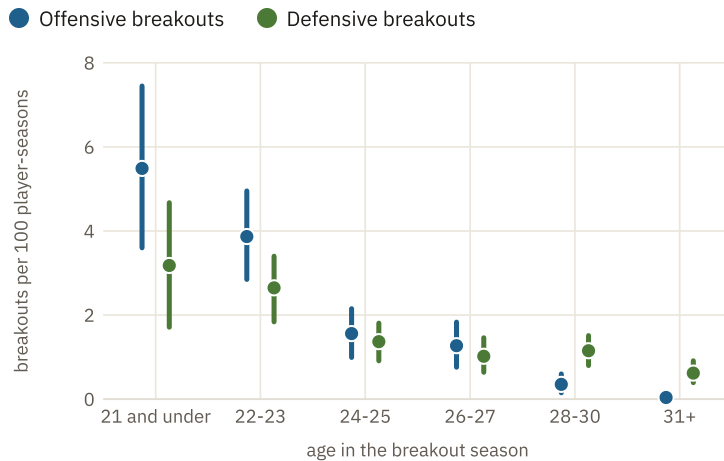
A skill-based game loses less through the late 20s, but not past the early 30s. From 27 to 31 the explosive group falls 0.20 league spreads and the skill group 0.07, a gap of -0.13 (-0.17 to -0.08); after 31 the two fall at the same pace.

Dot: estimate. Bar: 90% interval. Explosive group: rim share, dunks, transition share, free-throw rate, blocks, offensive rebounds. Skill group: three-point %, free-throw %, mid-range %, assists, ball security.

FIGURE 4**Offense rises fast when young and falls fast after 30; measured defense barely moves**

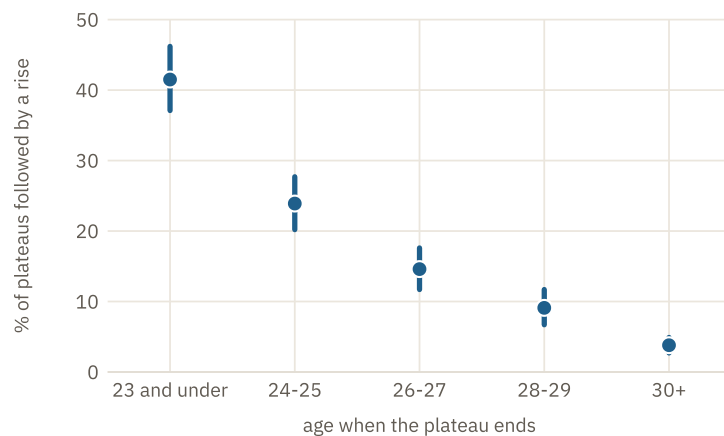
Defense is not the first thing to go, though the measure may be too blunt to see it go. Offensive impact rises 0.69 points per 100 a season at 22 and under and falls 0.46 a season at 33 and over. Measured defense moves far less: $+0.12$ when young and -0.03 when old.

Dot: average change from one season to the next. Bar: 90% interval. Only players who keep their minutes count, so late-career declines are understated.

FIGURE 5**Breakouts come young; defensive ones come about two years later than offensive ones**

Do not write off a young player's defense at 23. 52% of offensive breakouts come at 23 or under, against 33% of defensive ones (median age 23.9 against 25.7). Most of a breakout is not forecast: for an overall-impact breakout the forecast expected a median 18% of the rise, though the definition builds in some surprise. Young high picks are about five times over-represented among those breakouts, and a change of team is under-represented (0.53 times).

A breakout is a one-season rise in the top 2.5% against the player's own past. Dot: rate. Bar: 90% interval.

FIGURE 6**A plateau at 23 is often a pause; at 30 it is usually the level**

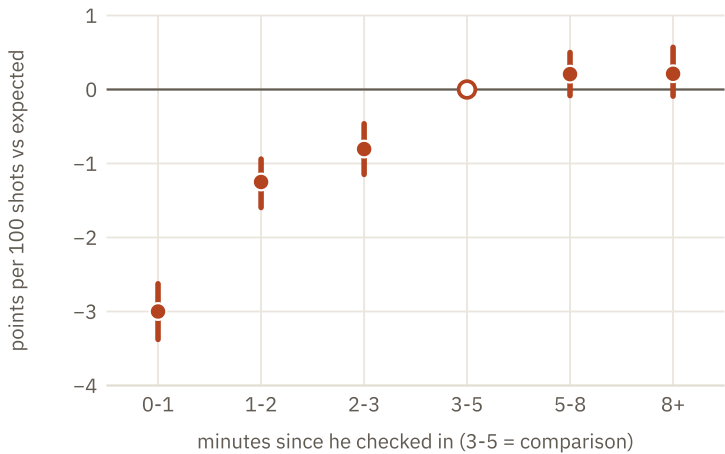
The chance of a rise after a plateau falls from 42% when it ends at 23 or under to 4% at 30 and over. Of 2,184 plateaus, 335 were followed by a rise larger than the forecast expected, often on another part of the game such as usage or creation.

Plateau: two or more flat seasons. Dot: share. Bar: 90% interval.

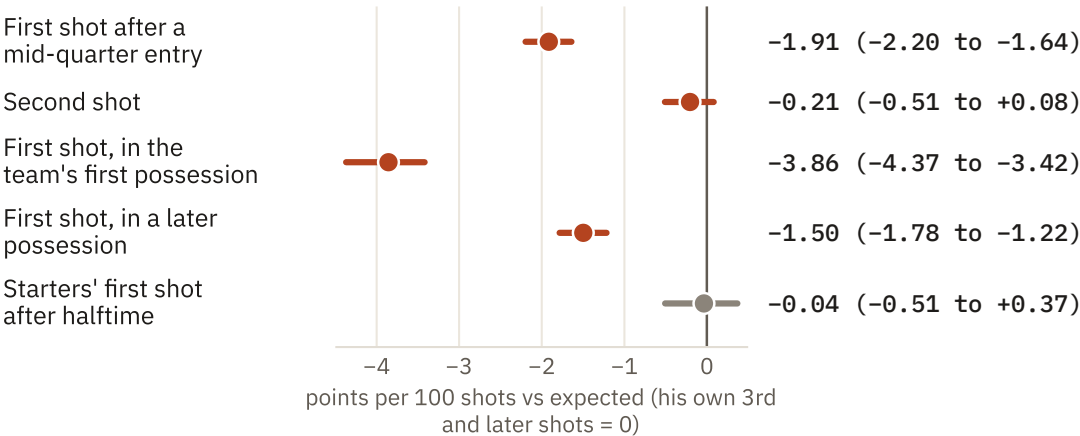
A related caution, for players 22 and under: a young player on a strong team can look worse than he is. Each extra point per 100 in his four teammates' combined prior-season rating lowers his measured impact by 0.125 per 100 (0.078 to 0.168).

FIGURE 7
A player's first shot after checking in is his worst, especially in the team's first possession

A. By minutes since entry



B. By shot order and possession



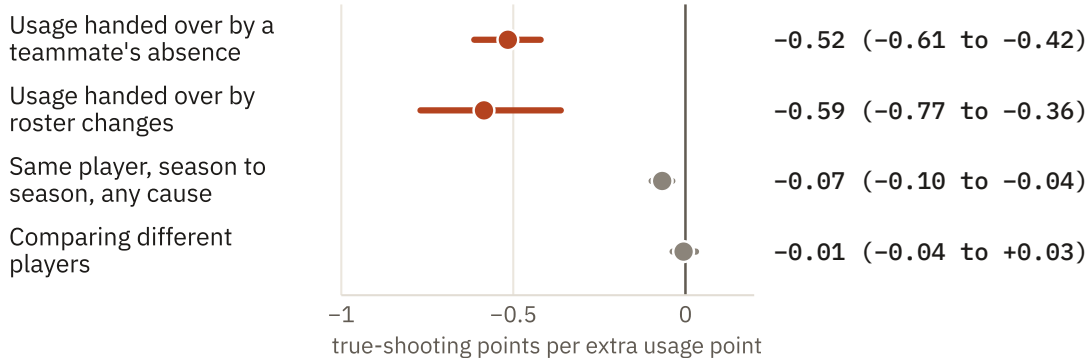
A player's first shot after a mid-quarter entry is worth 1.91 fewer points per 100 shots than his own later shots, after accounting for shot type, location and his skill. The first minute is worst (about 3 points), and the dip is gone after about three minutes. It is worst in the team's first possession after he enters (-3.86), against -1.50 in a later possession. The effect is small: about 0.3 points a game for a team. Moving that first-possession shot to someone else could recover 0.06 to 0.10 a game. Whether to act on it is the coaching staff's call.

Every regular-season shot 1996-97 to 2025-26. Dot: estimate. Bar: 90% interval. The recovery is 0.06 a game if his first shot moves to a later possession, and 0.10 if an average shot replaces it. The possession split is not tested on fresh data, and coaches may call plays for players they expect to be guarded poorly. Starters coming out of halftime, after a warm-up, show no dip. A player's own dip does not repeat from season to season, so it is no use for rating players.

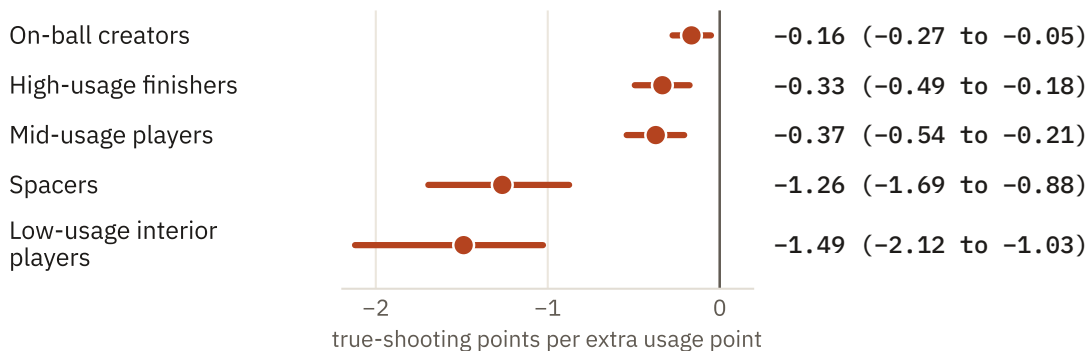
FIGURE 8

When a teammate's absence hands a player more of the ball, his true shooting falls, mostly from losing the teammate who set him up

A. The cost, by how it is measured



B. Who pays it (teammate-absence test)



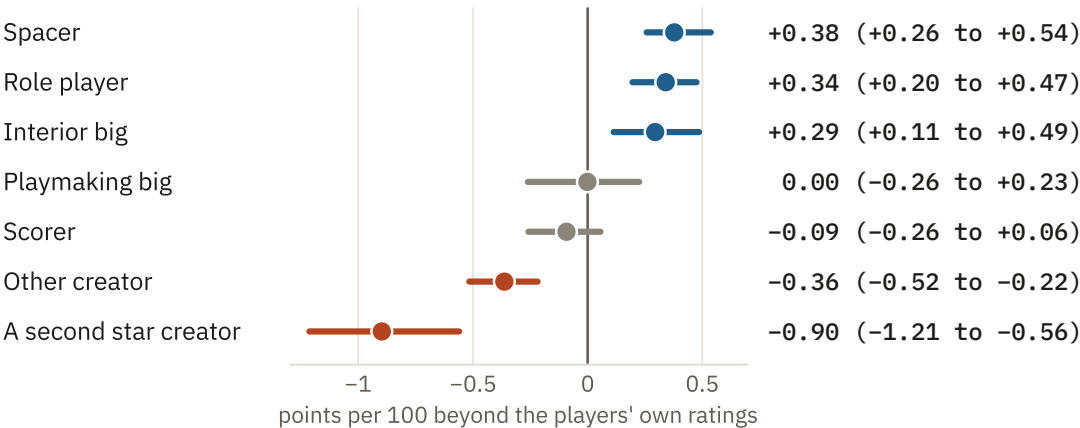
When a teammate's absence or a roster change hands a player more of the ball, his true shooting falls 0.52 to 0.59 points per usage point. Comparing players shows almost no cost, because coaches give the ball to the players who can carry it. In the teammate-absence test, most of the cost is losing the teammate who set him up; with that passing held fixed, the extra shots cost little: -0.06 (-0.21 to $+0.08$). The roster-change test cannot split the cost this way. On-ball creators pay little; spacers and interior players pay a lot. Usage is a resource to fit together as each player's efficiency develops.

Dot: estimate. Bar: 90% interval. Teammate-absence test: 1996-97 to 2025-26. Roster-change test: 2022-23 to 2025-26.

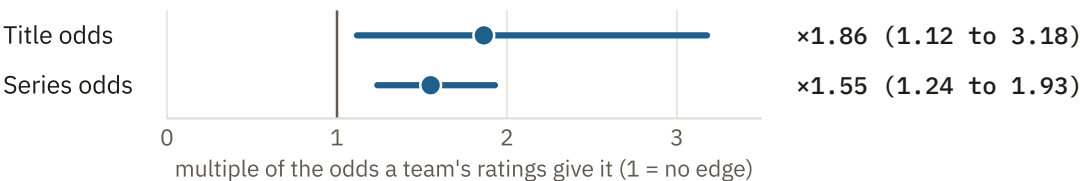
Fit: building around a star creator

FIGURE 9
Spacers and interior bigs fit next to a star creator; two star creators are worth less than the sum of their ratings

A. Next to a star creator, each type adds



B. Title and series odds with a lead option who shoots at least league average



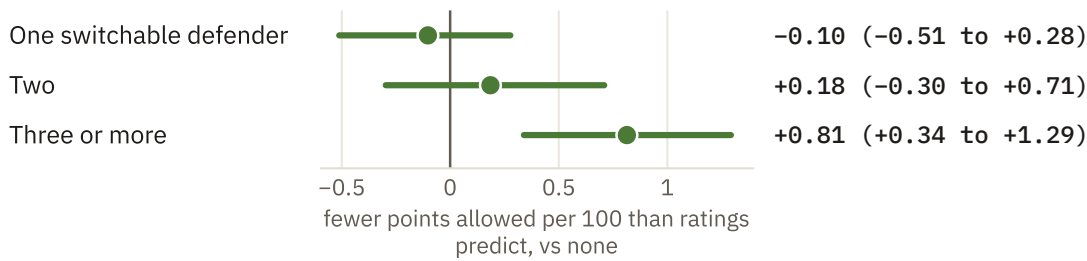
Next to a star creator, spacers add 0.38 points per 100 beyond their own ratings and interior bigs 0.29; two star creators together are worth 0.9 per 100 less than their ratings add up to. Pair effects held up in later seasons at 1.19× their size (0.69 to 1.69). Teams with a lead option who shoots at least league average won 32 titles; their ratings and playoff brackets predicted 26.7. As a group they were already favorites, so multiplying their odds by 1.86 adds about five titles over 46 seasons. Without the shooting condition the multiplier is ×1.46 (0.85 to 2.67), which cannot be told apart from no edge. A forward test on the 2008-26 playoffs was too small to confirm the gain.

Star creator (panel A): usage at least one standard deviation above average and offense in the season's top 10%. Lead option (panel B): a regular-minutes player whose league usage rank is at most half the number of teams (15th with 30 teams) and whose true shooting is at or above league average. Dot: estimate. Bar: 90% interval. Pairs: lineups 1996-97 to 2025-26. Titles: playoff teams 1980-81 to 2025-26.

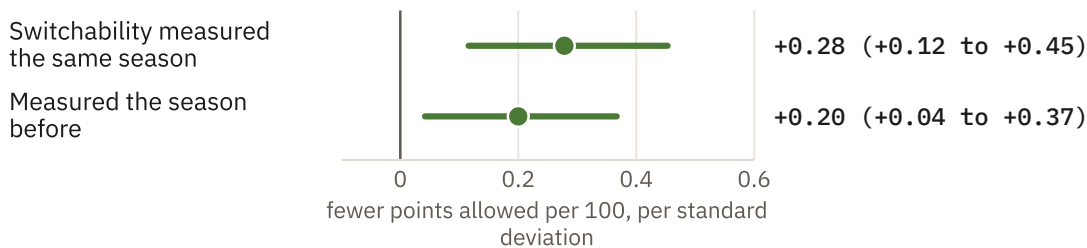
FIGURE 10

Lineups with more switchable defenders allow fewer points than their ratings predict

A. By number of switchable defenders



B. Per standard deviation of lineup switchability



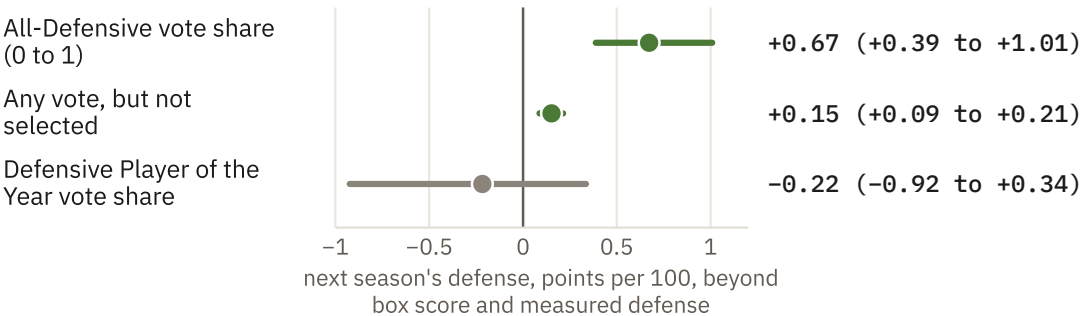
Most of the gain comes with the third switchable defender. Per standard deviation of switchability, lineups allow 0.28 fewer points per 100 than the five players' defensive ratings predict (0.12 to 0.45), and last season's switchability still predicts it. The intervals are wide.

Switchability: how widely the heights of the players a defender guards vary, beyond what his usual assignment predicts; a stable trait (reliability 0.72). NBA.com matchup data 2017-18 to 2024-25. Dot: estimate. Bar: 90% interval.

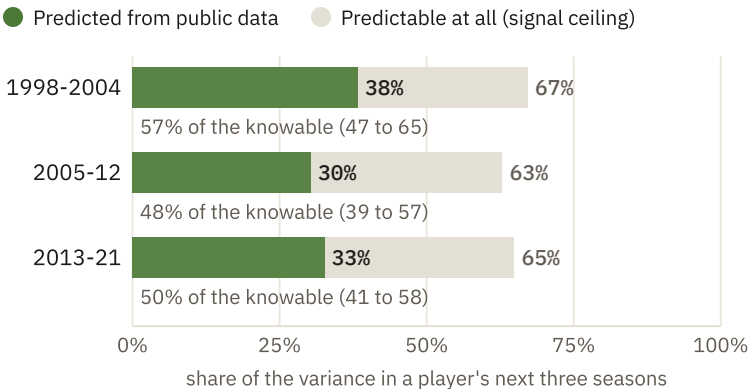
Defense, and the limits of public data

FIGURE 11
Award votes add defensive signal the numbers miss; public data captures about half of the signal in a player's future

A. What award votes add beyond box score and measured defense



B. How much public data can know

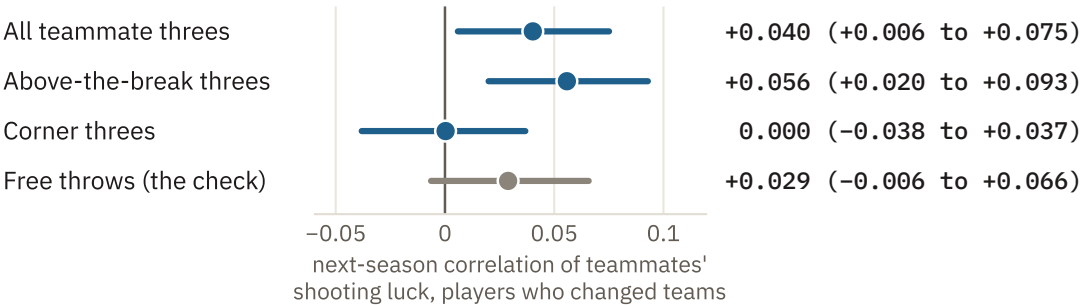


All-Defensive votes predict a player's defense next season beyond his box score and measured defense: +0.67 per 100 for a full share, and +0.15 for any vote short of selection. Public data explains 30 to 38% of a player's next three seasons of impact, and 63 to 67% of that impact is signal rather than noise, so public data captures about half of the signal. Better models of public data can close some of the gap; the rest needs data teams hold: tracking, medical, practice and scouting. In the draft, when draft order and a public stats model disagree, draft order wins 56% of the time on four-year value (51 to 62%).

A: 1996-97 to 2024-25; dot: estimate, bar: 95% interval. B: RAPM averaged over the next three seasons; the ceiling is the share of that average that is signal rather than noise, and it has no interval.

Central questions, tested

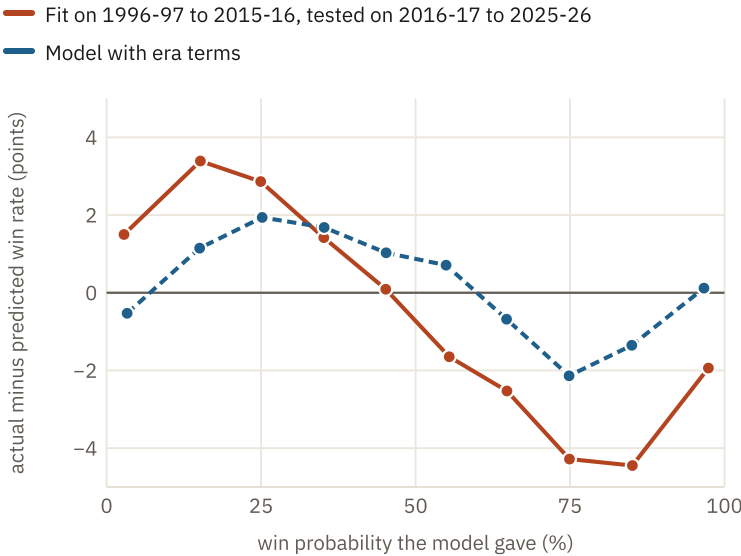
FIGURE 12
Gravity: a small effect, clearest on above-the-break threes



The evidence for gravity is suggestive, not settled. When a player changes teams, his new teammates' threes beat their norms in the same direction they did on his old team: weakly for all threes, more clearly above the break, not in the corners. The all-threes correlation (0.040) barely clears what random teammates produce by chance (0.038), and the free-throw check, where gravity should not show, is smaller (+0.029) but not zero, which weakens the result. A typical player's true effect is about 0.47 percentage points on each teammate three, but one season of the measure is about 98% noise. Several seasons, or tracking data, would settle it.

Dot: correlation. Bar: 90% interval. Free throws are the check because nobody guards them.

FIGURE 13
Leads are less safe than they used to be: an older win-probability model is overconfident on recent games

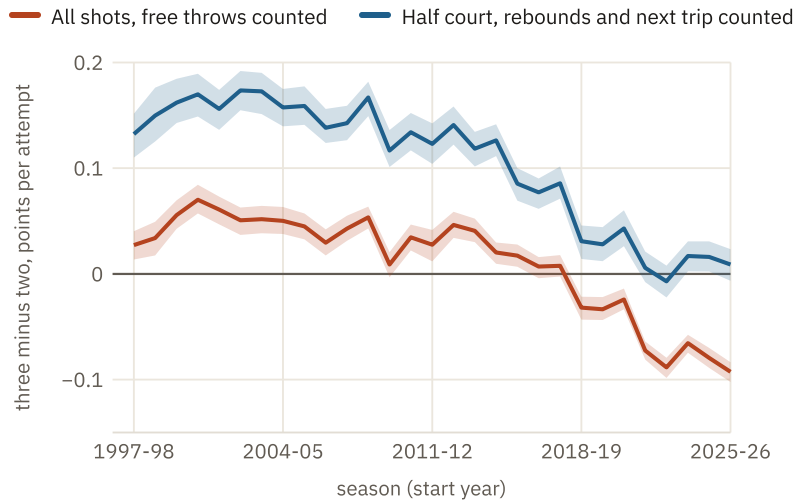


Leads are less safe than they used to be, so late-game decisions should use a model fit on the current era. A model built from score, clock and possession on 1996-97 to 2015-16 games is too confident on newer games: when it gave a team 85%, that team won 81%. Its odds need pulling back toward even by about a sixth (a slope of 0.83, where 1.00 means no correction). The model with era terms is closer but still off by about 2 points: when it gave a team 25%, that team won 27%, and when it gave 75%, it won 73%. Each season on its line is scored by a model fit on the other seasons, so it has already seen the current era. Each point is one of ten bins of predicted win probability. 0 = perfectly calibrated; below 0 on the right means favorites won less often than the model said.

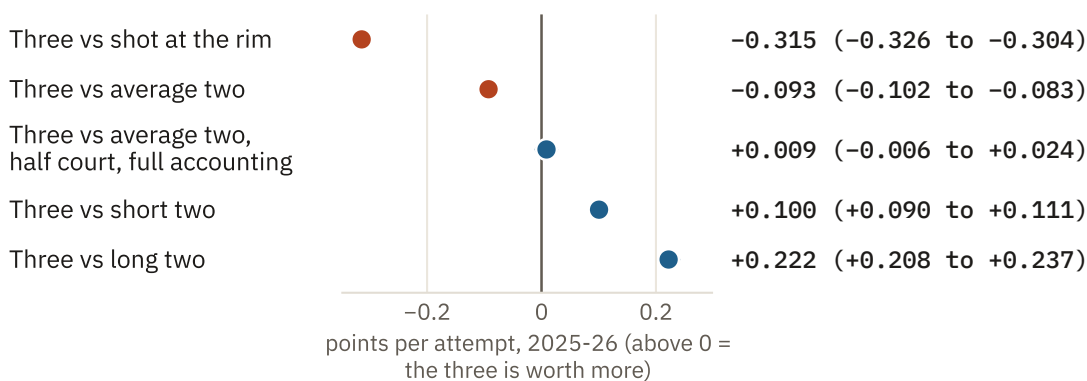
FIGURE 14

Threes versus twos: in the half court, fully counted, they are even, and the balance can be tracked every season

A. Three minus two, every season



B. By kind of two, 2025-26



In the half court, once offensive rebounds and the opponent's next trip are counted, threes and twos are even (+0.009, -0.006 to +0.024): NBA teams have found the balance. Across all shots, counting free throws but not rebounds, the average two returns 0.09 points per attempt more than the average three, so it can look as if teams take too many threes; leaving out putbacks and fast breaks removes about 60% of that lead. Shots at the rim are the only twos worth more than a three: get to the rim, and do not trade threes for midrange twos.

Band and bar: 90% interval. Half court excludes transition, putbacks, heaves and garbage time.

3 How ideas get tested

591

questions tested

221

answered no

57

mixed

313

answered yes

- Forecasts are scored on seasons the model never saw.
- Most numbers carry a 90% interval.
- With this many tests, some "yes" answers will be luck. Weigh single results with that in mind.

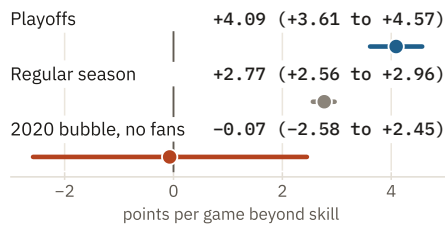
FIGURE 15

I thought to test it: eleven common beliefs, measured

These are beliefs widely held around the game. Most held up; what is new is their size, which a coaching staff can prepare with. Dot: estimate. Bar: 90% interval (95% for playing hurt). No bar: the test reported a single estimate or a spread.

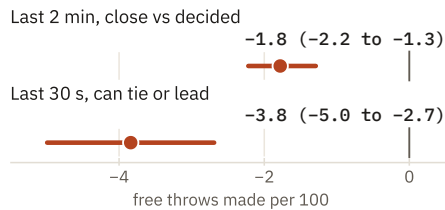
Playoff home court

Worth about 4 points a game beyond skill, 1.5 times the regular season's; it vanished in the 2020 bubble.



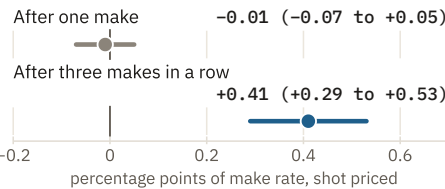
Free throws under pressure

Players make fewer free throws when the game is on the line.



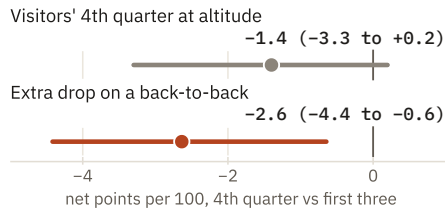
The hot hand

Real but small once shot difficulty is priced: hot players take harder shots.



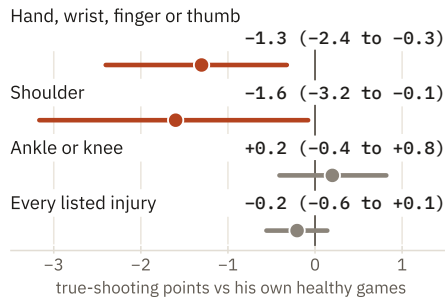
Altitude in Denver and Utah

Their home edges are the league's largest. Visitors' fourth-quarter fade is unclear on its own but shows on the second night of a back-to-back.



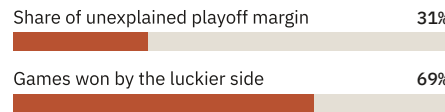
Playing hurt

Hand and shoulder injuries cut true shooting (about 1.3 and 1.6 points); ankle and knee injuries, and the average listing, do not move it.



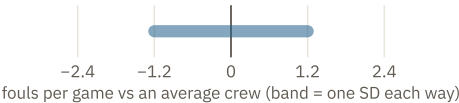
Luck in playoff surprises

Three-point luck explains about a third of what regular-season margins miss, though a real playoff shooting skill would also count as luck here.



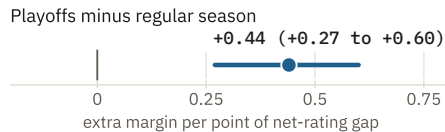
Referees

Crews differ by about 1.2 fouls a game (one SD). Each referee's tendency carries into the next season, so it can be known before tip-off.



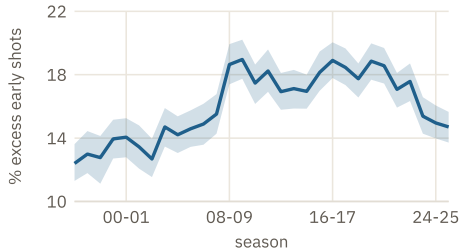
Playoffs widen quality gaps

Between playoff teams, a point of net-rating gap buys 1.19 points of playoff margin against 0.75 in the regular season.



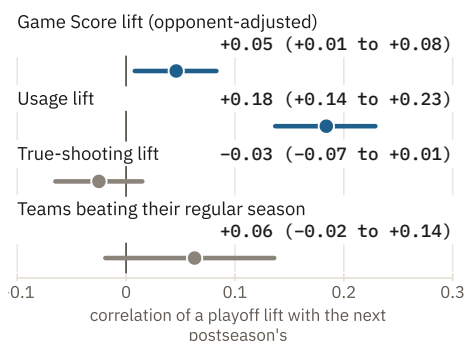
The two-for-one has faded

Quarter-end two-for-ones rose through the 2000s, jumped in 2008-09 and stayed high until 2022-23; since then teams go for them less. Why is not yet tested.



Playoff risers

Playoff stars are mostly regular-season stars (correlation 0.83). Some players repeatably take on more scoring in the playoffs and hold their production better than others, but the edge is small, it is not in shooting efficiency, and one postseason's lift is about 4% player and 96% noise.



Back-to-backs cost more now

A back-to-back costs a team about 2.6 points of margin since 2021-22, up from 1.7.

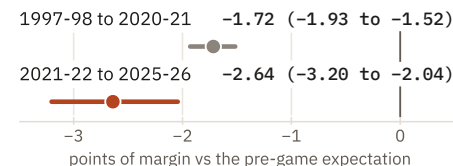


FIGURE 16

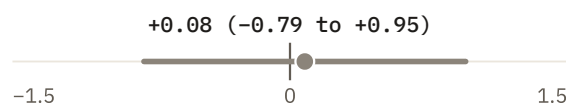
Tested and dropped: popular ideas the data does not support

Each idea sounds right and failed a fair test: the dot sits at or near no effect, or on the wrong side. Two point clearly the wrong way: a bigger role slightly lowers bench players' shot-making, and hunted defenders are not scored on more efficiently. Length still matters on defense: longer players defend better, even though taller guards do not.

Dot: estimate. Bar: 90% interval. Each row has its own scale; the vertical line marks no effect (equal say for pace).

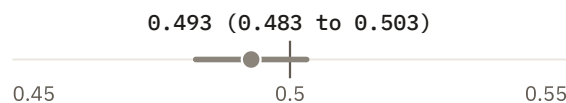
Teams facing elimination play above themselves

points per game



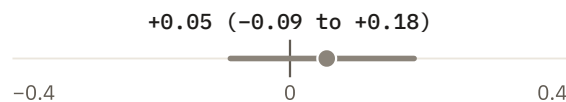
The faster team sets the pace

faster team's share of control (0.5 = equal say)



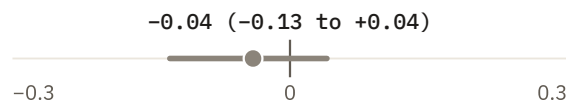
Being pulled off your own pace hurts your offense

points per 100 per 3 possessions of mismatch



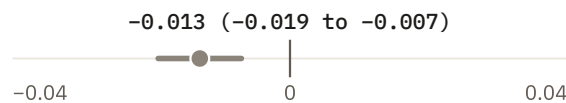
Taller guards defend better

defensive impact per 100 per inch of guard height



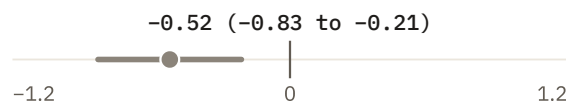
Playoff offenses score more per shot on the weak defenders they hunt

extra playoff points per attempt per SD weaker defense (a cost would be above 0)



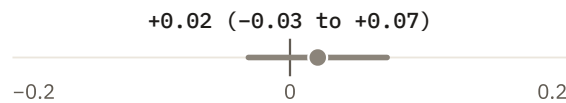
A bigger role makes a player shoot better

bench shot-making per 100 shots, per 10 points of extra role



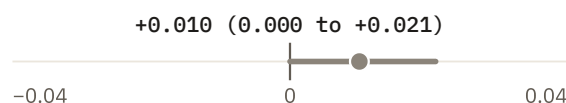
Clutch is a trait

late-and-close edge, next-season correlation

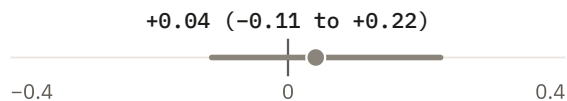


Stars get a reputation whistle

extra free throws per shot vs look-alikes after a first All-NBA team; at the edge of zero, and the shooting-foul rate does not move



Regular-season rest pays off in the playoffs
playoff true-shooting points per 3 fewer minutes a game



A college stat beats draft slot
0 of 15 out-of-sample tests

0 of 15

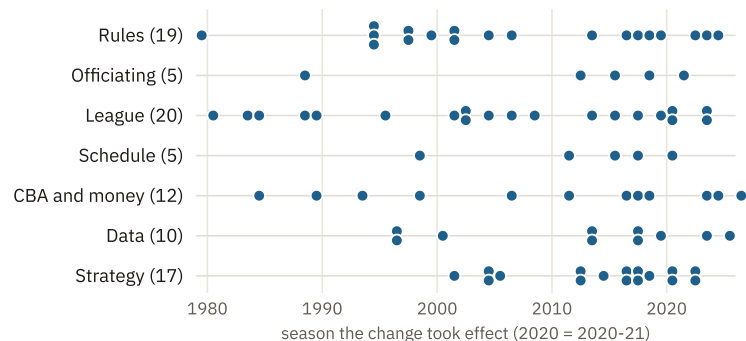
Style uniqueness predicts stardom
change in All-NBA forecast precision (no interval saved)



FIGURE 17

Qualitative history, quantified: every change to the game, dated

88 changes to the game (87 dated), 1979-80 to 2026-27



Any trend over time is read against what changed in its window: rule changes, officiating memos, CBA terms, data-feed changes and strategy shifts, each dated. On the web version, hover over a dot to see the change.

The idea the project is named after did not hold

Singular started from a hunch: the NBA, more than other sports, runs on singular talents, so a player whose style is unlike anyone else's should be likelier to become a star. Tested on future All-NBA players, style uniqueness slightly hurt the forecast, and a machine-learned style profile did worse.

That version of the idea could not be proven with public data alone. But the best players in the league plainly each have a unique game, so the nature of top talent remains a guiding principle. There is something there, and it is what I want to keep pursuing.

4 What I'm working on further

An analytics department can give the decision makers in the front office invaluable tools. Here are a few, each marked built or proposed.

PROPOSED

Player trajectory profiles. Each player's forecast for next season and beyond, split into shooting, creation, rim pressure, defense, minutes and value. Staff could set a custom path for one part (say a rookie's shooting improves faster than the model expects) and see what it does to the rest and to his long-term value.

BUILT

Cap and rules engine. The current CBA as code, with the league's announced 2026-27 cap, tax and apron figures and every team's cap position by season. It passes 103 of 113 cap-expert edge cases, from the stretch provision and apron trade matching to sign-and-trade hard caps. The other 10 are listed as known gaps.

PROPOSED

Trade-deadline scenario simulator. On that engine: forecast each team's likely moves and run thousands of league-wide deadline scenarios, updated as moves happen.

PROPOSED

Trade fit simulator. Score any trade by fit with the current core: switchable defenders, and spacers and interior bigs next to the star creator (Figures 9 and 10).

PROPOSED

Organization forecasting app. Anyone in the organization would rate players from a phone: underrated, overrated or fair. Each call would be scored as seasons play out, showing which signals and whose judgments predict best.

PROPOSED

More qualitative signals. Coaches' All-Star votes, Most Improved votes and scouting language, each held to the test the All-Defensive votes passed (Figure 11).

PROPOSED

Free-agent supply forecast. Who will reach free agency and when, from the rookie-extension and restricted-free-agency models already fitted.

PROPOSED

Automated briefings. Pre-game and event briefings for every part of the organization: schedule density, altitude back-to-backs, the referee crew's foul rate, and new data as it arrives.

PROPOSED

Defensive matchup ratings, effort and gravity from tracking. Score each defender against each attacker's own usual rates; measure effort as speed and distance on like-for-like plays; measure gravity directly from where defenders stand. Public data cannot separate these from role.

PROPOSED

A day-one plan for the team's tracking feed. A 15-step plan, already written: week one would check the feed against numbers reproduced from public data; month one would build what public data cannot.

5 Contact

Griffin Hilly

TERMS USED

Per 100 possessions	points scored or allowed per 100 trips down the floor, so fast and slow teams compare fairly.	Net rating	points scored minus points allowed per 100 possessions.
True shooting	shooting efficiency that counts threes and free throws.	Usage	the share of a team's plays a player finishes (shot, free throws or turnover) while he is on the floor.
RAPM	a player's impact measured from the lineups he played in, with teammates and opponents accounted for.	90% interval	the range the true value plausibly falls in; a bar that crosses zero means "cannot tell".
League spread	one standard deviation across players, so traits on different scales can share a chart.		

Prepared October 2026. Intervals are 90% unless a figure says otherwise.